

Idea 1 — Full Paper Design & Roadmap

*When the Regime Returns: Retention and Invariance in
Decision-Focused Strategy Pools for Non-Stationary Markets*

Yuhao Li (University of Pennsylvania)

Source assets: GAUSE (Li, 2026) + Inv-PnCO (Yan, Geng, ..., Li, et al., TPAMI 2026)

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1. Paper Title

Primary title: “*When the Regime Returns: Retention and Invariance in Decision-Focused Strategy Pools for Non-Stationary Markets.*”

Alternatives (pick at submission based on venue culture):

- (a) “Invariant Regime Specialists: Reward-Independent Capital Allocation for Decision-Focused Learning under Regime Switching”
- (b) “Two Kinds of Non-Stationarity: Decomposing Retention and Invariance in Regime-Switching Strategy Pools”
- (c) “RISP: Regime-Invariant Specialist Pools — Surviving Both Dormancy and Drift in Financial Decision-Making”

Framework name (use throughout paper and code): RISP (*Regime-Invariant Specialist Pools*).

2. Thesis: The Single Falsifiable Claim

Financial markets exhibit **two orthogonal kinds of non-stationarity** that existing strategy-allocation methods conflate:

(N1) Between-regime switching. Regimes (trend / range / high-vol / crisis) recur on irregular schedules with long dormancy. A reward-chasing capital allocator (recent-P&L weighting, learned MoE router) starves and overwrites the dormant-regime specialist, paying a full recalibration cost exactly when the regime returns [GAUSE Principle 1].

(N2) Within-regime drift. Each occurrence of a regime is a different draw from the regime’s environment distribution (the 2008 crisis $\neq$ 2020 $\neq$ 2022). A retained specialist trained by ERM on one episode still fails OOD on the next episode [Inv-PnCO motivation].

Thesis statement. A population of capacity-bounded strategy specialists, (i) assigned to regimes by winner-take-all competition so that the converged regime-to-specialist map is *reward-independent* (solving N1 — the crisis specialist idles through calm markets and retains its calibration by identity, not P&L), and (ii) each trained with a decision-focused *invariance* objective (mean + variance of SPO-style regret loss across historical episodes of its own regime, solving N2 — the specialist learns the invariant decision-oriented factors common to all episodes of its regime rather than overfitting one), achieves strictly lower post-reactivation decision regret than (a) recent-performance capital allocators, (b) capacity-matched regime-conditioned monoliths, (c) learned MoE routers, and (d) retained-but-ERM specialists — and the gain decomposes additively into a RETENTION term and an INVARIANCE term.

Scope statement (preempting the sleeping-experts deflation). The contribution concerns capacity-bounded *learners*, not combinations of fixed experts. Sleeping-experts / adaptive-regret schemes also freeze dormant experts’ state — but only because their experts are *static strategies* with nothing to forget. The problem RISP solves arises precisely when specialists must keep *learning* within their niche under bounded capacity, so that reward-driven reallocation causes destructive interference across regimes. State this scope in the introduction, or a knowledgeable online-learning reviewer deflates the paper in one paragraph.

The paper in one picture (Figure 1 of the paper). A 2×2 grid. Rows: allocation mechanism {reward-driven, reward-independent}. Columns: specialist training {ERM, invariant}. Cell value: post-reactivation regret. Prediction: monotone improvement along both axes, near-additive interaction. *This 2×2 is the contribution* — nobody has separated these two failure modes before.

3. Novelty Claims (intro contributions list, near-verbatim)

C1 — Problem decomposition. First work to formally decompose non-stationary financial decision error into a between-regime *forgetting* term and a within-regime *invariance gap*, and to show they are addressed by independent mechanisms (allocation vs. training objective).

C2 — Method. RISP: the first strategy-pool architecture combining reward-independent (emergent, competition-driven) capacity assignment with decision-focused invariant learning per niche. No gate to train, no freezing schedule, no hand-set regime-to-desk map.

C3 — Theory. A post-reactivation regret bound of the form

$$\text{Regret}_{\text{react}} \leq G_{\text{inv}}(\beta, \text{episode heterogeneity}) + G_{\text{forget}}(D, \text{allocation mechanism}),$$

where $G_{\text{forget}} = 0$ for any reward-independent assignment (extends GAUSE Observation 1 to the PnO/regret setting) and G_{inv} is controlled by the Inv-PnCO variance penalty (adapts Inv-PnCO Theorem 1 to regime-indexed environment families). See §7.

C4 — Evaluation protocol. A new evaluation standard for regime-aware financial ML: *post-reactivation regret* measured in a probe window of the first N trading days after each regime reactivation, with leakage-free regime labeling, deflated performance statistics, and a pre-registered structure diagnostic (GAUSE §8.6) run *before* claiming gains.

C5 — Honest audit. Conditions under which RISP does *not* pay: unlimited capacity, no regime structure, regimes that never go dormant — plus the financial analogue: when carrying costs of idle specialists exceed reactivation savings. Quantify the *break-even dormancy length*.

4. Positioning & Related-Work Map

Bucket A — Regime-switching finance. Hamilton (1989) Markov-switching; Ang & Timmermann (2012); Guidolin & Timmermann; recent ML regime-aware portfolios, HMM+RL hybrids, statistical jump models (Nystrup et al.). *Differentiation:* these detect regimes and condition *one* model; none address bounded capacity + dormancy-driven forgetting of the specialist itself. Our monolith arm is this literature’s best case.

Bucket B — Decision-focused learning / PnO. SPO (Elmachtoub & Grigas), SPO-relax (Mandi et al.), LODL, LTR-PnO, Inv-PnCO. *Differentiation:* all assume stationary or one-shot OOD environments; none handle *recurring* environments with dormancy under bounded predictor capacity.

Bucket C — Online learning / ensembles. Hedge/MWU over strategies; universal portfolios; online portfolio selection (Li & Hoi); **adaptive regret / sleeping experts is the closest**

formal relative — must be addressed head-on. Crucial nuance: sleeping experts *also* retain dormant experts’ state, but only because its experts are fixed strategies with nothing to forget. The differentiation is the scope statement of §2: RISP addresses capacity-bounded *learning* specialists, where reward-driven allocation causes cross-regime interference. Accordingly, baseline A8 ships in *two* variants: (A8a) sleeping experts over fixed strategies (expected to retain but underperform within-regime, since it cannot adapt), and (A8b) sleeping experts over capacity-bounded learning experts (expected to inherit the interference/forgetting failure). This pair turns the most dangerous related work into a supporting experiment.

Bucket D — MoE / continual learning. Li et al. (ICLR 2025) MoE-CL theory (gate freezing); EWC. Used exactly as in GAUSE: corroborating theory, complementary regime.

Bucket E — Multi-strategy fund practice (motivation). Strategy-decay literature (McLean & Pontiff), crisis-alpha / tail-hedge literature. Grounds the “crisis playbooks decay during calm” claim of GAUSE §9.2.

5. Formal Problem Setup (Paper §3)

Data stream. $t = 1, \dots, T$; observable features x_t (returns, vol, term structure, breadth, flows, ...), unknown coefficients y_t (next-period returns/costs), latent regime $r_t \in \{1, \dots, R\}$. Regimes recur; dormancy $D(r)$ is the length of inactive spells.

Episodes as environments. The e -th historical occurrence of regime r defines environment (r, e) with distribution $p(x, y \mid r, e)$. **Assumption** (Inv-PnCO Assumption 1, regime-indexed): within each regime r there exist invariant decision-oriented factors f_r such that $p(z \mid f_r, e) = p(z \mid f_r)$ across episodes e — the *optimal decision structure* of a crisis is stable across crises even though surface features drift (flight-to-quality correlations, vol clustering, liquidity spirals recur structurally).

Episode-scarcity mitigation (statistical viability of the variance penalty). Crisis regimes yield only 5–7 historical episodes in 35 years of equities; $\text{Var}_e[\mathcal{L}]$ over 5 noisy samples is fragile. Two mitigations are part of the *default* design, not afterthoughts: (i) *sub-episode environments* — bootstrapped non-overlapping blocks within episodes enlarge the environment set while preserving between-episode heterogeneity (block length sensitivity reported); (ii) *cross-asset episodes* — the same crisis observed in equities, credit, FX, and commodities counts as distinct environments. Mitigation (ii) is a feature, not a patch: it upgrades the claim to *invariance across asset classes within a regime*, which is both stronger and more interesting than within-asset invariance, and it directly exploits the multi-asset data plan of E6.

Decision layer (kept combinatorial to inherit PnCO):

- *Primary*: cardinality-constrained long-only portfolio

$$\max_z y_t^\top z \quad \text{s.t.} \quad \sum_i z_i \leq \text{budget}, \quad \|z\|_0 \leq k, \quad z \in \{0, w_{\min}, \dots, w_{\max}\} \text{ (grid),}$$

knapsack-like, solvable with Gurobi; regret well-defined as $\text{Regret}(t) = |F(z^*, y_t) - F(\hat{z}, y_t)|$ [Inv-PnCO Eq. 4].

- *Secondary (robustness)*: (i) asset-selection knapsack; (ii) optimal execution scheduling as shortest path (symmetry with Inv-PnCO’s SP task).

Population. N specialists, each with capacity $K < R$ (a specialist keeps $\leq K$ regimes calibrated). Implement both GAUSE memory models: hard LRU on regime-specific heads, and soft interference decay via per-regime adapters that decay when untouched.

RISP mechanism (batched competition cycle).

1. All specialists produce predictions $\hat{y}_i$ and decisions $\hat{z}_i$ for the current regime r_t .
2. Score: quality $q_i = -\text{Regret}_i$ (normalized); competitive score $q_i + \lambda(\alpha_{i,r_t} - 1/R)$.
3. **Batched winner-take-all (default).** Per-period financial regret is noise-dominated (daily return signal-to-noise is tiny), so per-step competition risks converging to noise. The winner is therefore decided over rolling competition windows of $W_c \approx 20$ trading days of *cumulative* within-regime regret; the window winner applies one EG affinity update ($\eta \approx 0.6$). The per-step variant is retained as an ablation, with a W_c sweep (1, 5, 20, 60 days) characterizing the noise-responsiveness trade-off. (This is the financial-data adaptation GAUSE’s cleaner prediction-error domains did not need.)
4. Specialists holding r_t within capacity update their predictive model for r_t with the Inv-PnCO loss over that regime’s episode buffer:

$$\min_e \text{Var}_e[\mathcal{L}_{\text{SPO}}(r_t)] + \beta \mathbb{E}_e[\mathcal{L}_{\text{SPO}}(r_t)],$$

where e ranges over stored episodes of r_t (bounded episode memory counts toward K).

5. After convergence (affinities > 0.95): *pin* the assignment (GAUSE deployment recipe, step 5); re-open competition only if the regime inventory changes (structure-diagnostic drift).

Regime labeling (critical; three causal, leakage-free schemes; report all, headline under L1): L1 volatility-band + trend-sign rule; L2 HMM fit on past data only with online filtered states; L3 statistical jump model.

6. Experiment Design (Paper §5)

6.1 E0 — Structure diagnostic (pre-registered gate, runs first)

GAUSE §8.6 oracle test per asset class: does regime-conditioning with true labels beat shuffled labels *on the decision-regret metric*? Only passing asset classes enter the headline; failures are reported. Note: GAUSE found crypto/commodities lack exploitable structure under the *prediction* metric — but prediction-optimal $\neq$ decision-optimal (the core PnO lesson), so structure may reappear under regret. This re-test is itself a publishable finding.

6.2 E1 — Headline: the 2×2 dissociation (post-reactivation regret)

Arms (capacity-matched, same method/feature inventory):

Arm	Description	Allocation	Training
A1	Monolith (capacity K , regime-conditioned)	reward-driven	ERM
A2	MoE router (gate trained on realized regret)	reward-driven	ERM
A3	Recent-Sharpe capital allocator (12m lookback; <i>industry baseline</i>)	reward-driven	ERM
A4	Random fixed niches	reward-indep.	ERM
A5	RISP-ERM (competition allocation, ERM training)	reward-indep.	ERM
A6	RISP-full (competition + Inv-PnCO loss)	reward-indep.	INV
A7	Monolith + Inv-PnCO loss	reward-driven	INV
A8a	Sleeping experts over <i>fixed</i> strategies	adaptive	none
A8b	Sleeping experts over capacity-bounded <i>learning</i> experts	adaptive	ERM

Note on A8a/A8b: this pair operationalizes the scope statement of §2 — A8a is expected to retain but underperform within-regime (fixed experts cannot adapt); A8b is expected to inherit the interference failure. Together they localize RISP’s advantage to the bounded-learner setting.

Metrics: overall regret; *post-reactivation regret* (first $N=15$ days after each reactivation, mirroring the GAUSE probe window); net-of-cost Sharpe in probe windows; max drawdown around transitions; relearning half-life.

Pre-registered predictions (the falsifiable core): post-reactivation regret $A6 < A5 \approx A4$ (with coverage gaps) $\ll A2 \approx A3 \approx A1$; A7 beats A1 overall but still fails post-reactivation (invariance cannot rescue a forgotten specialist); the A6–A5 gap grows with episode heterogeneity.

6.3 E2–E5 — Sweeps and ablations

- **E2 Capacity sweep:** $K = 1, \dots, R$, both memory models. Expect GAUSE Fig. 7 shape: reward-independent arms flat in K ; reward-driven close only as $K \rightarrow R$.
- **E3 Dormancy-length sweep (semi-synthetic):** regime-stitched bootstrap of real per-regime blocks into controlled schedules, $D \in \{1m, 6m, 1y, 3y, 5y\}$. Yields the *break-even dormancy length* for the cost audit (C5).
- **E4 Episode-heterogeneity sweep** (isolates the invariance axis): hold allocation fixed at RISP; vary training episodes per regime and inject controlled covariate shift between episodes. ERM degrades with heterogeneity; invariant specialist flat (mirrors Inv-PnCO Fig. 9c).
- **E5 Ablations:** $\lambda = 0$ and λ sweep; β sweep; *without* variance term (expect the Inv-PnCO Table-7 failure: worse than ERM); competition-window sweep $W_c \in \{1, 5, 20, 60\}$ days (per-step vs. batched); episodes-as-environments vs. random-split environments; pinned vs. never-pinned; decision-layer swap (portfolio / knapsack / execution-SP); L1 vs. L2 vs. L3 labels.

6.4 E6 — Data

- **Crypto:** Bybit daily (have; 8,766 bars) + top-20 universe for cross-sectional portfolios.
- **Commodities:** FRED (have) + futures continuous series.
- **Equities (flagship):** S&P 500 constituents daily, 1990–2025 (CRSP via WRDS if available; else free survivorship-aware sources). Covers 2000, 2008, 2011, 2015, 2018, 2020, 2022 stress episodes $\Rightarrow$ 5–7 crisis “environments” — enough episodes per regime for the invariance loss.
- **FX (optional):** G10 daily.

Walk-forward protocol: expanding window, strict $t+1$ execution, embargo gaps at regime boundaries to kill label leakage.

6.5 E7 — Statistical rigor (preempting the backtest-overfitting attack)

20–30 seeds per stochastic component; 95% CIs; Welch tests with Holm–Bonferroni (the GAUSE protocol); Deflated Sharpe Ratio (Bailey & López de Prado) for any Sharpe claim; Hansen SPA / White Reality Check across the arm family; transaction costs at 5/10/25 bps plus borrow costs (net-of-cost everywhere); idle-specialist carrying-cost model for break-even analysis; *no* hyperparameter tuned on the test period (nested walk-forward for β, η, λ).

7. Theory Component (Paper §4) — Proposition Sketches

Prop. 1 (Regret decomposition). Under the regime-indexed invariance assumption, for a pool with assignment map A and per-specialist hypothesis q , expected probe-window regret after reactivation of regime r with dormancy D satisfies

$$\mathbb{E}[\text{Regret}_{\text{react}}(r)] \leq C_1 \sqrt{\text{Var}_e[\mathcal{L}_e(q_r)]} + C_2 \Phi(D; A),$$

(invariance gap + forgetting deficit). *Proof plan:* term 1 by adapting Inv-PnCO Lemma 2 + Theorem 1 (episodes = environments; test episode = e'); term 2 by GAUSE Observation 1 (hard model: $\Phi = \Pr[\geq K \text{ distinct regimes hit specialist during } D] \times \text{full relearn cost}$; soft model: $\Phi = (1 - (1 - \rho)^D) \times \text{cost}$).

Prop. 2 (Reward-independence zeroes forgetting). If A is independent of the realized reward process (pinned converged competition, random fixed, frozen gate), the specialist for r receives zero foreign updates during dormancy and $\Phi(D; A) = 0$ for all D (hard model; soft model: bounded by the regime’s own drift). Reward-driven A has $\Phi \rightarrow \text{full cost}$ as D grows.

Prop. 3 (Interaction). The two terms are minimized by independent design choices; corollary: the 2×2 of E1 is predicted monotone in both axes.

Scope honesty: idealized-model results (as in GAUSE); do *not* claim adversarial-regret guarantees for the coupled winner-take-all dynamics.

Known technical gap (budget time for it). Inv-PnCO Theorem 1 bounds a *KL divergence* between solution distributions, not regret. Bridging $KL \rightarrow \text{regret}$ in Prop. 1 requires an additional step (bounded objective values + a Pinsker-type inequality, with constants depending on the objective range). This is standard but not free: budget +2 weeks of theory time, and keep the scoped-down fallback of stating Prop. 1 in the KL metric (still sufficient for the decomposition message) if the regret-metric constants become ugly.

8. Anticipated Reviewer Attacks & Prepared Responses

R1 “Backtest overfitting / no real alpha.” We claim *regret reduction and retention*, not alpha discovery; DSR + SPA tests; semi-synthetic dormancy sweeps with known ground truth; the 2×2 dissociation is a mechanism claim robust to absolute P&L levels.

R2 “Regime labels arbitrary / lookahead.” Three causal labeling schemes; embargo at boundaries; shuffled-label control.

R3 “Just retrain quickly at reactivation.” The probe-window cost *is* the relearning cost; in crises the first 15 days are when relearning is most expensive. E3 quantifies it; adaptive baseline A8 included and loses in proportion to regime complexity.

R4 “Isn’t this sleeping experts / adaptive regret?” The scope statement (§2) + the A8a/A8b pair answer this constructively: sleeping experts over fixed strategies retains but cannot adapt within-regime (A8a); over capacity-bounded learning experts it inherits the interference failure (A8b). RISP’s contribution is localized to the bounded-learner setting, where adaptive-regret machinery does not help. Shown empirically, with a dedicated related-work paragraph.

R5 “Funds already silo desks by regime.” Yes — RISP shows the silo structure can *emerge*, and shows precisely when recent-P&L capital allocation *across* silos (which funds do use) destroys the siloing benefit. The target is the allocator, not the org chart.

R6 “Invariance assumption untestable in finance.” E4 tests its consequences; the without-variance ablation shows the assumption does work; stated with the same status as Inv-PnCO Assumption 1.

R7 “Capacity bound K is artificial for software strategies.” The soft interference model (shared backbone + decaying per-regime adapters) reflects real parameter interference; the ops-cost framing (calibration staleness, infra, attention) is industry reality (GAUSE §9.2). Both models give the same dissociation.

9. Fallback Plans (decision tree)

- F1** If E0 finds no exploitable regime structure on the decision metric in any real asset class $\Rightarrow$ pivot headline to semi-synthetic regime-stitched markets + real data as “structure diagnostic in the wild”; retitle toward “... A Controlled Study”; the diagnostic-first methodology becomes a co-equal contribution (still ICAIF/AAAI-publishable).
- F2** If retention is real but invariance adds nothing ($A6 \approx A5$) $\Rightarrow$ the 2×2 with a null cell is still a finding (“retention dominates; invariance does not transfer to the allocation problem”); shrink invariance to an ablation; expand theory + dormancy sweeps; consider merging with the theory-companion paper.
- F3** If invariance helps but retention is weak on real data (dormancy too short in sample) $\Rightarrow$ pivot application to *risk-model committees* (GAUSE §9.2(iii)): retention measured on stress-VaR calibration error after long calm stretches — dormancy is years long and measurable without P&L. Venue shifts toward risk journals + ICAIF.
- F4** If the decision layer bottlenecks $\Rightarrow$ drop to pure knapsack on a 50-asset universe (Inv-PnCO scale ran on one RTX 2080).
- F5** If reviewers want LLM-era relevance $\Rightarrow$ one future-work paragraph bridging to Idea 3 (RISP allocating memory across LLM analyst agents). Do not let it grow.

10. Roadmap & Timeline

Primary target (staggered plan): AAAI-27 / KDD-27 winter cycle, with ICLR 2027 (~Sept 2026; verify exact dates) as the stretch option decided Aug 1. Backups: ICAIF-26 workshop track for visibility; AISTATS-27 (Oct).

Phase	Window	Milestones
1	Jun 10–24	Freeze formalization in a 4-page tech note; port GAUSE population loop; add Gurobi decision layer + regret metrics; assemble equities panel (submit WRDS request <i>now</i>); implement labelers L1–L3 with causality tests; run E0. [GATE: E0 go/no-go]
2	Jun 24–Jul 15	Implement 8 arms; replicate GAUSE-style retention on financial data (sanity); implement Inv-PnCO loss with episodes-as-environments (reuse <i>hygeng</i> /PT0 SPO machinery); first full E1 on equities. [GATE: 2×2 ordering; if wrong, diagnose ≤ 2 weeks then trigger F1/F2/F3]
3	Jul 15–Aug 19	E2/E3/E4 sweeps; E5 ablations (incl. W_c competition-window sweep); write Props. 1–3 with proofs, <i>including the $KL \rightarrow \text{regret}$ bridge</i> (+2 weeks budgeted; decide by Jul 1 whether to invite an Inv-PnCO coauthor for theory review); statistical battery (DSR, SPA, costs).
4	Aug 19–Sept 9	Figures (Fig. 1 2×2 ; Fig. 2 architecture; Fig. 3 capacity sweep; Fig. 4 dormancy sweep; Fig. 5 retention timeline with real crisis dates); full draft; red-team against §8 attack list.
5	Sept 9–deadline	Reproducibility package; timestamped pre-registration appendix of E1 predictions; submit to the winter cycle (KDD-27 research / AAAI-27 if dates align; the +2-week theory buffer is absorbed by the staggered plan).

Post-submission: workshop version to ICAIF-26 / NeurIPS-26 workshops; journal long version to *Quantitative Finance* or *Journal of Financial Data Science*.

11. Resources & Dependencies

- **Compute:** modest — Inv-PnCO ran on one RTX 2080; GAUSE is light. One workstation or \$200–500 cloud. Gurobi academic license (free with Penn email).
- **Data:** Yahoo/Stooq for prototyping; CRSP via Wharton WRDS for the flagship equity panel (request early; lead time is the risk).
- **Code reuse:** HowardLiYH/GAUSE + hygeng/PTO (branch Rethink1.0).
- **Authorship:** solo-feasible; recommended to invite one Inv-PnCO coauthor for theory credibility (decide early — affects framing).

12. Field Classification

Primary field: Machine Learning for Quantitative Finance / Decision-Focused Learning (PnO).

Secondary: Multi-agent learning; continual learning; online learning.

Venues: ICLR / AAAI / KDD (ML framing); ICAIF (finance framing). Journals: *Management Science*, *Quantitative Finance*, *JFDS*.

Keywords: decision-focused learning; regime switching; catastrophic forgetting; invariant learning; portfolio optimization; emergent specialization; non-stationarity.

13. Abstract Draft (v0 — refine after E1)

Financial markets are non-stationary in two distinct ways: regimes switch and recur on irregular schedules, and each recurrence of a regime is distributionally different from the last. We show that these two kinds of non-stationarity defeat existing strategy-allocation methods through two independent mechanisms — reward-chasing capital allocation forgets dormant-regime expertise, and ERM-trained specialists fail out-of-distribution within their own regime — and that each is solved by a different design choice. We propose RISP (Regime-Invariant Specialist Pools), in which capacity-bounded strategy specialists acquire regimes through winner-take-all competition, yielding a converged regime-to-specialist assignment that is independent of current P&L (so crisis specialists idle through calm markets and retain calibration), while each specialist trains with a decision-focused invariance objective across historical episodes of its regime (so retained specialists generalize to the next occurrence, not the last one). We prove a post-reactivation regret bound that decomposes into a forgetting term — zero for any reward-independent assignment — and an invariance gap controlled by an episode-variance penalty, and we verify the predicted 2×2 dissociation across equities, crypto, and commodities under three leakage-free regime labelings, with deflated performance statistics and explicit transaction-cost and break-even-dormancy analyses. An honest audit maps exactly when the mechanism pays: bounded capacity, recurring dormancy, and verifiable regime structure — a diagnostic we run before, not after, claiming gains.